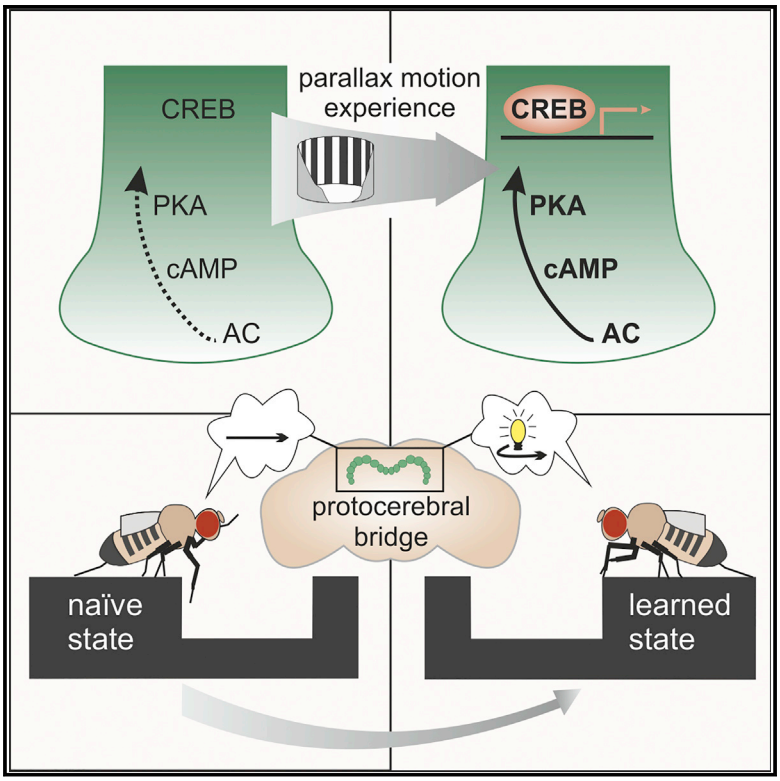
Report

Current Biology

Drosophila Acquires a Long-Lasting Body-Size Memory from Visual Feedback

Graphical Abstract



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In Brief

Observing the decisions of a fly to surmount gaps exceeding its body reach, Krause et al. show that naive flies vastly overestimate their reach. However, walking in a striped arena suffices to establish a long-lasting body-size memory. Intrinsic neurons of the protocerebral bridge require dCREB2 activity for memory consolidation and maintenance.

Highlights

- Fruit flies build a most stable long-term memory for their body size and body reach
- Body-size memory is acquired through parallax-motion experience while walking
- Memory is consolidated after 2 h but cannot be retrieved earlier than 12 h
- Memory consolidation and long-term maintenance depend on dCREB2 activity

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Drosophila Acquires a Long-Lasting Body-Size Memory from Visual Feedback

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SUMMARY

Grasping an object or crossing a trench requires the integration of information on the operating distance of our limbs with precise distance estimation. The reach of our hands and step size of our legs are learned by the visual feedback we get during our actions. This implicit knowledge of our peripersonal space is first acquired during infancy but will be continuously updated throughout our whole life [1]. In contrast, body size of holometabolous insects does not change after metamorphosis; nevertheless, they do have to learn their body reaches at least once. The body size of *Drosophila* imagines can vary by about 15% depending on environmental factors like food quality and temperature [2]. To investigate how flies acquire knowledge about and memorize their body size, we studied their decisions to either refrain from or initiate climbing over gaps exceeding their body size [3]. Naive (dark-reared) flies overestimate their size and have to learn it from the parallax motion of the retinal images of objects in their environment while walking. Naive flies can be trained in a striped arena and manipulated to underestimate their size, but once consolidated, this memory seems to last for a lifetime. Consolidation of this memory is stress sensitive only in the first 2 h after training but cannot be retrieved for the next 12 h. We have identified a set of intrinsic, lateral neurons of the protocerebral bridge of the central complex [4, 5] that depend on dCREB2 transcriptional activity for long-term memory consolidation and maintenance.

RESULTS AND DISCUSSION

Flies attempting to cross a gap wider than their step size show stereotypical leg-over-head searching movements of the front legs that initiate a climbing attempt [3] (Figures 1A and 1B). However, if the gap is definitively too wide to be surmounted, the approaching fly will turn away and does not initiate this typical behavior. Interestingly, smaller flies turn away at gaps that larger individuals still attempt to cross, indicating that individual flies have knowledge about their own body size and the operating distance of their legs [6]. To investigate this in further detail, we confronted cohorts of small and large wild-type flies from the same set of culture vials with increasing gap sizes. We classified the flies visually and measured characteristic leg lengths, which are positively correlated with their body size [7], after the behavioral experiments (Table S1). When confronted with a gap size that just exceeds their body size (3.0 mm), small flies avoided crossing in 50% of their approaches and increasingly either turned around or walked down the gap without an attempt, whereas larger flies showed similar changes in behavior at the 3.5-mm gap (Figures 1C–1E). The leg-over-head behavior is the best predictor for an attempt to cross a gap that cannot be easily surmounted by a large step (see Videos S1 and S2), and small flies started to show this behavior at the 2.5-mm gap (Figure 1F). In contrast, their on average 15.8% larger siblings started to do so at the 3.0-mm gap. As judged by the leg-over-head behavior, small flies showed reduced climbing efforts at the insurmountable 4.0-mm gap and almost never attempted to cross at a width of 4.5 mm or wider. In contrast, the larger siblings made about 20% more attempts at the 4.5-mm gap, and even at 5.5 mm an attempt to cross was made in about 10% of the approaches (Figure 1F). This suggests that flies do have a knowledge about their body size and adjust their behavior to their individual reach. Because leg-over-head behavior is the best measure for the intention to cross an insurmountable gap, we used this criterion (percentage of attempts within 10 approaches to the gap) for the rest of the study.

Flies Learn Their Body Size through Visual Feedback

We hypothesized that flies would need to connect visual input with tactile experience; therefore, we expected that dark-reared (DD) flies would be less able to judge their chances of crossing a gap than controls reared in circadian light-dark (LD) cycles. Indeed, 3-day-old flies that had been deprived of visual information from pupal development onward attempted to climb an insurmountable gap of 5.0 mm with a median frequency of 50% compared to 0% of LD controls (Figure 2A). This was not due to a lack of visual acuity, because they were able to cross 3.5-mm gaps as successfully as the controls (Figure 2B). This is in line with the normal optomotor responses of DD flies [8], despite the fact that dark rearing reduces the volume of the first optic neuropil [9]. To determine whether DD flies could learn their body reach during the ten climbing

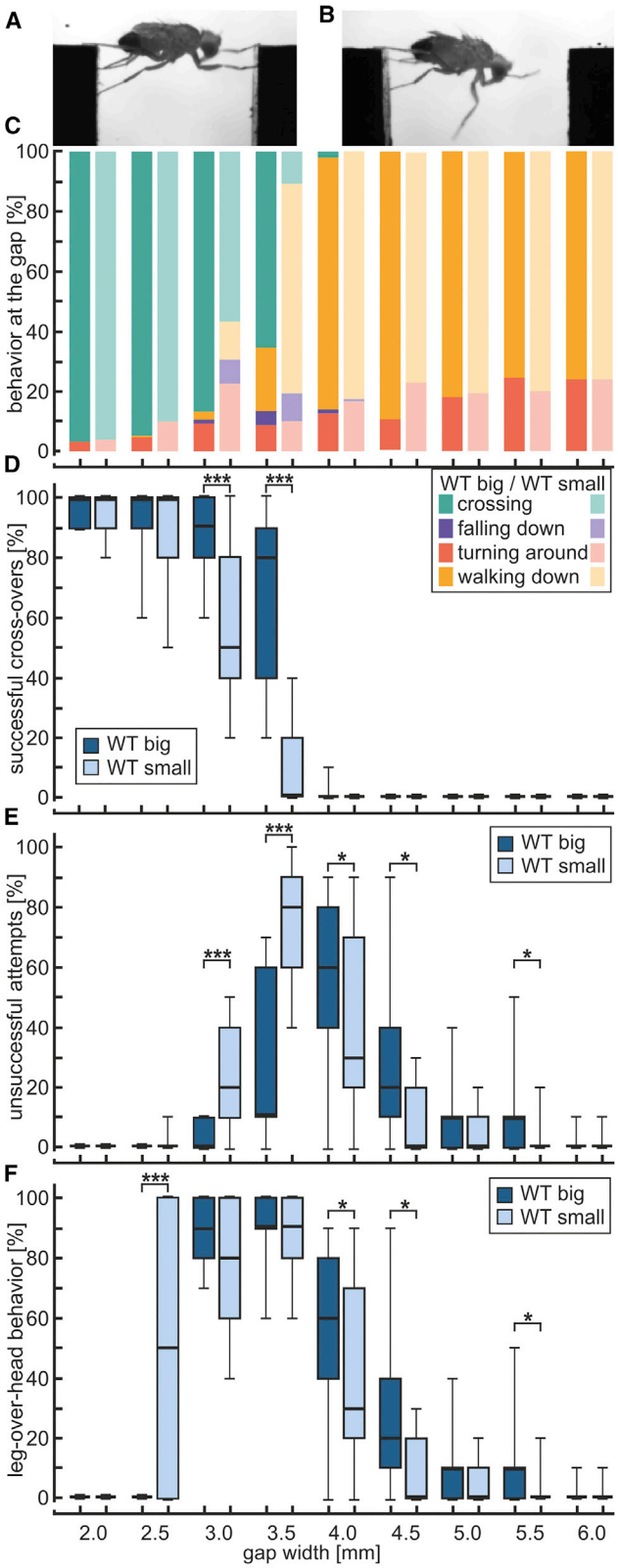

Figure 1. Body Size Determines the Number of Crossing Attempts in Normal Flies

(A) Snapshot of a wild-type fly crossing a 3.5-mm gap, which exceeds its body size but not its body reach.

(B) A wild-type fly attempting to cross an insurmountable 4.5-mm gap as indicated by typical leg-over-head movements.

(C) Summary of the observed behaviors (see also Videos S1 and S2) of big and small flies (N = 15; 10 approaches per fly) at different gap sizes (2.0–6.0 mm). Wild-type Canton-S flies from the same vial were visually sorted for especially small or big flies and separated into two groups. After the behavioral test, the tibia plus tarsi length was evaluated as a proxy for body size (average size difference, 15.8%; see Table S1 for leg-length measurements.

(D–F) Statistical comparison of climbing success (D), unsuccessful attempts (E), and observed leg-over-head behavior (F) of big and small flies at increasing gap size. Boxes represent 25% and 75% quartiles; whiskers represent 5% and 95% quantiles; *$p < 0.05$, ***$p < 0.001$; Shapiro-Wilk's test for normal distribution; Mann-Whitney U or Student's t test for independent groups.

attempts of the test (approximately 3 min in the light), we put them back in the dark and tested them again at 5.0-mm gaps the next day. Although they performed significantly better in the second test, the rate of futile attempts was still higher than in normally reared flies. This incomplete knowledge about their body reach was not due to the lack of successful climbing experience, because the outcome is not different when flies were trained on a surmountable gap of 3.5 mm (Figure 2A). This suggests that the short visual input and the training at the gap during the first test only partially established knowledge about their body reach.

Next, we asked whether physical interactions with other flies during same-sex aggression [10] provide the fly with the visual and tactile information to establish its peripersonal space. Therefore, we isolated individual males during the pupal phase and let them hatch in small vials under LD conditions but without visual contrasts (a plain white paper shielded each vial), either alone or with two other males, before testing them 3 days after eclosion. Isolated flies were not able to estimate their body reach and tried to climb the 5.0-mm gap in 30% of their approaches, a frequency only slightly lower than measured in naive DD animals (Figure 2C; compare to DD in Figure 2A). Surprisingly, grouped males were not significantly different from isolated males (Figure 2C). We therefore asked whether flies need more specific visual information from their environment to determine their body size and reach. To address this question, single pupae were isolated as before, but now the vials were wrapped with a pattern of horizontal black and white stripes each of 5-mm width. Although being isolated and only instructed by the striped pattern, flies later refrained from climbing attempts ("stripes" in Figure 2C), similar to normally raised control animals. We therefore concluded that the flies learned to estimate distance and body reach by connecting the visual input from the stripes with their own body size while walking across the striped pattern. Walking generates parallax motion of the retinal images of structures in the environment. The average step size of an individual, which is proportional to its leg length and therefore to its body size [7], will create an average parallax motion. It has been shown earlier that flies determine gap width by the parallax motion caused by the opposite wall during an approach [3]. This neuronal circuit is obviously calibrated after eclosion to match individual body size.

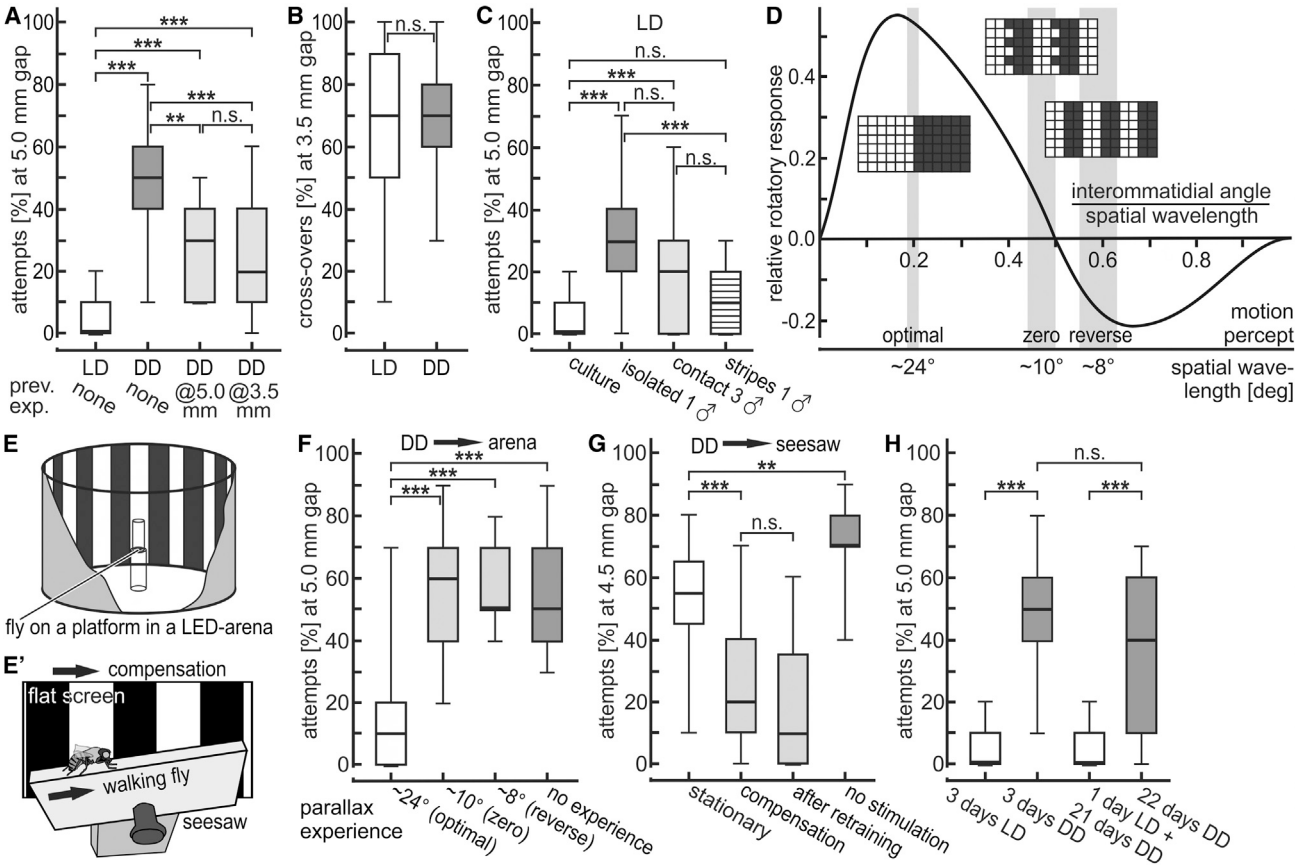

Figure 2. Flies Form Permanent Body-Size Memory from Parallax-Motion Experience

(A) Flies reared under normal light-dark (LD) conditions avoid futile attempts at a 5.0-mm gap, in contrast to dark-reared (DD) flies. Groups of DD flies that previously experienced ten approaches to an insurmountable 5.0-mm or surmountable 3.5-mm gap showed fewer attempts at the 5.0-mm gap when tested the following day, but both groups still differ from flies with experience in culture vials under LD. N = 30 flies per group. All boxplots display the median percentage of climbing attempts from ten approaches per Canton-S (CS) fly at an insurmountable 5.0-mm gap. Boxes denote 25% and 75% quartiles; whiskers denote 5% and 95% quantiles; n.s., not significant; *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$; Mann-Whitney U test or Student's t test for independent groups; Student's t test for dependent groups; corrections for multiple testing.

(B) DD flies are as successful in crossing a surmountable 3.5-mm gap as the LD control group; N = 15; Wilcoxon test for independent groups.

(C) Compared to control flies from culture vials, flies kept in small vials with a white cover under LD conditions since their pupal stages, either in isolation (isolated 1♂) or with two other males (contact 3♂), show excessive climbing attempts. In contrast, LD flies kept in isolation but in a vial covered with black and white horizontal stripes (stripes 1♂) learn their body size and perform like flies from culture vials. N = 30 flies per group; Kruskal-Wallis test; post hoc correction for multiple testing.

(D) Sensitivity of the motion-vision system [11] is almost optimal for patterns of 24° spatial wavelength. Due to interference between eye raster and pattern, the sensitivity is close to zero for 10°, whereas 8° causes an apparent reversal of motion direction (geometric interference). Flies were trained in an LED arena (E) with 2° × 2° pixels (insets). Viewing angles varied for the flies, which walked freely within a 10-mm radius in the center, as indicated by the gray zones.

(E) Setup for visual experience training. A single fly can freely move on a platform of 10-mm radius in between two vials in the center of an LED arena. The wall of the vial is made inaccessible. During training, either one of the stationary patterns (shown in D) is presented on the cylindrical LED screen.

(E′) Seesaw training apparatus exploiting negative geotaxis of flies as a driving force for continuous walking. Flat screens on either side were used to partially compensate the parallax motion resulting from walking by about 50% (see also Video S3) or to show a stationary striped pattern.

(F) Climbing attempts of DD flies, exposed 24 h beforehand for 10 min to one of three striped patterns (explained in D) in an arena as shown in (E). Only the optimally trained group (~24°) avoids attempts at the insurmountable 5.0-mm gap. The flies trained with zero (~10°), reverse (~8°), or no pattern (dark reared until testing) try to climb in 50%–60% of the approaches. N = 15 flies per group; Mann-Whitney U test for independent groups; corrections for multiple testing.

(G) DD flies were given 15 min of visual experience with partial compensation of their parallax motion in the training apparatus (seesaw in E′) and tested the next day. For comparison, another group of DD flies was trained in this apparatus, but with a stationary, non-compensating pattern. The partially compensated flies show significantly fewer trials than the stationary trained flies. Retraining of the partially compensated flies for 15 min with a stationary pattern the next day did not change the number of climbing attempts when tested the following day. All trained groups show significantly fewer attempts at the 4.5-mm gap than the DD flies without visual stimulation. N = 16 flies per group; Student's t test for independent groups; Wilcoxon test for dependent groups (compensation versus retrained); corrections for multiple testing.

(H) Flies reared under normal LD conditions do not forget their body size after 21 days of darkness. They avoid attempts like 3-day-old flies staying in normal LD. Performance of naive flies reared in the dark for 22 days functions as a control for overall climbing activity. They initiate climbing in 40% of the approaches, not significantly different from 3-day-old naive flies taken out of the dark. N = 15 flies per group; Kruskal-Wallis test; post hoc correction for multiple testing.

Flies Learn Their Body Size through Parallax Motion

The motion-vision system of the fly can be manipulated by specific regular patterns interfering with the regular eye raster of the compound eyes. A regular striped cylinder surrounding the fly can lead to annihilation of the motion stimulus or inversion of the perceived direction [11]. In our case, the pattern is stationary, whereas the fly is moving. Annihilation is given, e.g., for a spatial pattern wavelength of twice the interommatidial angle of 4.8°, whereas the inverted direction is perceived for spatial wavelengths between the interommatidial angle and twice this angle (geometric interference creates reverse motion for a fly; "∼8°" in Figure 2D). On the other hand, the stimulation is strongest (and in the expected direction) when pattern wavelength is about 5 to 6 times the interommatidial angle ("∼24°") [11]. To test whether parallax motion can either instruct or confuse a fly about its body size, DD flies were exposed at day 3 after eclosion to one of the patterns. Single DD flies with wings shortened were allowed to walk, and perceive parallax motion, for 10 min on the bottom of an empty polystyrene vial of 20-mm diameter in the center of a light-emitting diode (LED) panorama (Figure 2E). The wall of the vial was coated with Sigmacote to prevent walking there, which would create other perspectives of the striped patterns for the visual system. 20 mm allows for about ten steps in a row; changing viewpoints vary the perceived pattern wavelength, as indicated by the gray zones in Figure 2D; note that the sensitivity curve of the motion-vision system is taken from flight experiments [11]. Indeed, when tested 24 h later at a 5.0-mm-wide gap, the flies that had experienced the 24° stimulus avoided climbing attempts ("optimal" in Figure 2F), whereas flies that were exposed to 10° ("zero") or 8° ("reverse") pattern wavelengths tried as hard as DD flies with no visual experience ("no experience" in Figure 2F).

The experiments described above suggested that flies learn their body size from integrating the visual information obtained from parallax motion and their average step length. If so, artificially reducing parallax motion during training should give a naive fly the impression of a smaller body size. Therefore, we put naive flies on a seesaw (because they like to walk upward) between two monitors showing vertical striped patterns (Figure 2E′; Video S3) and tested them the next day at a gap width of 4.5 mm where small flies, in contrast to larger ones, mostly refrain from climbing. If the stripes were kept stationary, the flies perceived a natural visual feedback from their walking activity and learned their body size. Those flies performed 50% of climbing attempts the next day at a just-insurmountable gap size of 4.5 mm. In contrast, reducing parallax motion during learning by half (by moving the stripes at half the average walking speed of the flies) tricked flies to perform only 20% of climbing attempts the next day at the 4.5-mm gap, showing that they underestimated their body size (Figure 2G). Even when we retrained these flies with the correct parallax motion the next day, they did not learn their real body size anymore, indicating that the initial information has been consolidated in the 24 h before retraining (see below).

Body-Size Memory Is Extremely Long Lasting

As mentioned above, for a holometabolous insect it should suffice to learn its size once and then to consolidate this memory for the rest of its life. To test this hypothesis, we subjected experienced (3 days in LD) flies to 21 days of darkness. Even after this long time, the flies remembered their body size and showed the same number of climbing attempts as 3-day-old control flies. Age did not affect the decisions to climb, because 22-day-old DD flies took the same high number of futile attempts as 3-day-old DD flies (Figure 2H). We therefore conclude that body-size memory is the most enduring memory described in *Drosophila* so far.

Consolidation of olfactory memory (both aversive and reward learning) into middle-term or long-term memory in *Drosophila* [12] can be impaired by cold anesthesia [13–15] or by induced elevated locomotion and sleep prevention [16, 17]. Therefore, we asked whether body-size memory could also be affected by these interventions. Applying cold shock (1 min on ice) immediately after optimal training (in an LED arena showing 24° pattern wavelength as in Figure 2F) did not significantly change the number of climbing attempts the next day (Figure 3A). In contrast, placing trained flies immediately in an empty vial on a nutator shaker and letting them roll overnight, which induces constant locomotion and sleeplessness, did prevent memory consolidation. When we started this agitation 2 h after training, the body-size memory was as good as in untreated flies (Figure 3A), indicating that memory consolidation was completed at that time and does not require sleep and/or rest for fixation. This short period of consolidation raised the question of at which time point after learning the body-size memory can be retrieved. In contrast to associative olfactory learning [12, 18], optimally trained flies were not able to retrieve body-size memory 3 min after training (Figure 3B), and neither could flies retrieve it 4 h after training, notably a time point when consolidation is irreversible by stress. First significant retrieval was observed as late as 12 h after training, and perfect memory was observed after 14 h.

Next, we wondered whether there is a sensitive imprinting period after eclosion, in which neurons that are not terminally differentiated are yet able to form and consolidate body-size knowledge. To address this question, we compared the climbing decisions of optimally trained 3- and 9-day-old DD flies. Both groups learned their body size to the same extent, excluding a critical period in which young flies can learn but aged flies cannot (Figure 3C).

Lastly, we asked whether *de novo* protein synthesis is required during memory consolidation and/or for memory maintenance later on. It has been shown that feeding the protein synthesis inhibitor cycloheximide (CXM; 35 mM) for 1 h before olfactory conditioning affects long-term memory formation in flies, whereas application of CXM 24 h after the training did not change the memory score [19]. We performed a similar experiment by feeding 35 mM CXM for 2 h right after the training in a striped arena or 24 h after we had instructed the flies with parallax motion (Figures 3D and 3E). Naive (DD) flies that received CXM 3 min after training were not able to consolidate a body-size memory (Figure 3D). In contrast to findings on associative olfactory long-term memory [19], feeding CXM for 2 h after a consolidation period of 1 day did affect the maintenance of body-size memory (Figure 3E). Thus, dissimilar to the associative long-term memory, protein synthesis is still required a day after consolidation.

Body-Size Memory Formation Requires cAMP Signaling in Intrinsic Neurons of the Protocerebral Bridge

We previously showed that the central complex (CX) is involved in visual targeting during gap climbing and orientation behavior

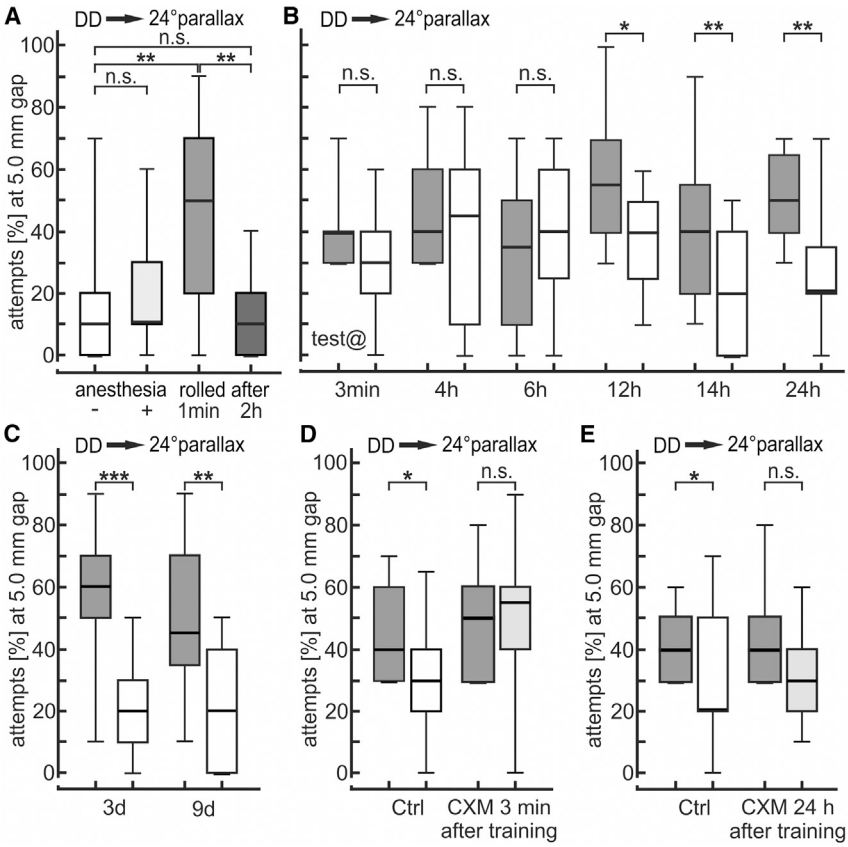

Figure 3. Timeline of Memory Consolidation and Retrieval

(A) Immediately after optimal training in the LED arena, DD flies were subjected to cold anesthesia for 1 min, which did not affect body-size memory the next day. Flies that were rolling in a vial on a nutator shaker 1 min after training until the next morning exhibit excessive climbing attempts. In contrast, flies treated 2 h after training until the next day avoid unsuccessful attempts not significantly different from undisturbed individuals. N = 15 flies per group; thick lines denote medians, boxes denote 25% and 75% quartiles; whiskers denote 5% and 95% quantiles; n.s., not significant; *p < 0.05, **p < 0.01, ***p < 0.001; Kruskal-Wallis test; post hoc correction for multiple testing.

(B) Groups of DD flies were tested for their naive phenotype (gray boxes) and subsequently trained in an optimally striped arena. Tests were performed after 3 min up to 24 h later (white boxes). Memory retrieval was observed only after 12, 14, and 24 h (white boxes). N for naive/trained: 3 min (15/15), 4 h (14/14), 6 h (24/24), 12 h (12/12), 14 h (24/22), and 24 h (12/12). Wilcoxon test for independent groups.

(C) DD flies were optimally trained in a striped arena when 3 (N = 14) or 9 days old (N = 10) and tested the next day (white boxes) at a 5.0-mm gap. DD flies of the same age (gray boxes; 3 days, N = 10; 9 days, N = 12) served as age-matched controls; t test for independent groups.

(D) DD flies were tested for their naive phenotype (gray boxes) and subsequently trained in an optimally striped arena. 35 mM cycloheximide (CXM) or 10% glucose, 3% ethanol control solution was fed for 2 h in constant darkness (DD) shortly (3 min) after training. CXM-treated flies showed no memory when tested the next day (light gray). N for naive/trained: 23/20 (control) and 21/18 (CXM). Wilcoxon test for independent groups; corrections for multiple testing.

(E) Feeding 35 mM CXM 24 h after training for 2 h in constant darkness (DD) also affected body-size memory. N for naive/trained: 26/19 (control) and 25/21 (CXM). Wilcoxon test for independent groups; corrections for multiple testing.

[20], suggesting that this brain region might be involved in establishing body-size memory. In Diptera, the CX consists of four neuropils: the protocerebral bridge (PB), fan-shaped body (FB), ellipsoid body (EB), and paired noduli (NOs) [4]. We also recently showed that *ocelliless* (oc^1) mutants, which lack most of the 18 glomeruli of the PB, are defective in visual targeting [20]. Testing oc^1 flies resulted in a generally high percentage of unsuccessful climbing attempts at insurmountable gaps, with small flies performing more leg-over-head behavior than large flies at 4.5- and 6.0-mm gaps (Figure S1). Therefore, an intact PB seems to be necessary to acquire a body-size memory.

Because our experiments suggested that the knowledge about body reach is gained through learning, we tested mutants that are impaired in learning and memory formation in our paradigm. The cyclic adenosine monophosphate (cAMP) signaling cascade contributes to different types of short- and long-term memories in *Drosophila*, and the dynamic changes of cAMP concentration in the flies' neurons are mainly driven by the synthesis through the adenylyl cyclase encoded by the *rutabaga* (*rut*) gene and metabolism via the phosphodiesterase encoded by the *dunce* (*dnc*) gene [21]. Therefore, we tested groups of small and large rut^1 and dnc^1 mutant flies (Table S1) for unsuccessful attempts at different gap sizes. Small rut^1 mutant flies displayed the same frequency of unsuccessful attempts at wide gaps as their larger siblings. A similar deficit in adaptation to actual body size was found in dnc^1 mutant flies (Figure S1).

These results, together with our finding that an intact PB might be required for learning and/or memorizing body size, suggested that a regulated cAMP signaling cascade is required in the PB. We therefore performed cDNA rescue experiments of rut^1 mutant flies [22] with GAL4 driver lines that express Rut in different subsets of PB neurons (Figures 4A and 4B), which are required for visual targeting at a gap [20]. 007Y-GAL4 drives expression in columnar neurons (PFNs), which form regular projections (including the so-called w, x, y, and z bundles) between the PB, FB, and NOs [23], whereas 210Y-GAL4 expresses in a different, latero-lateral set of PB (LPB) neurons (for a schematic drawing, see Figure 4A [20, 24]). As seen in Figure 4C, expression of Rut in the LPB neurons could rescue the body-size memory deficits of the rut^1 mutant. Because both GAL4 lines have broad expression pattern, we also tested two additional lines from the Janelia FlyLight collection [25] that drive in latero-lateral neurons (R55G08-GAL4; Figure 4F) and in columnar neurons of the PB (R73D06-GAL4 [26]). As seen in Figure 4C, expression of Rut only in R55G08 neurons could also rescue the body-size memory deficits of the rut^1 mutant. As visualized by synaptic markers [27], the R55G08-marked neurons (also called Δ7 neurons [28]) have pre- and postsynapses distributed through all glomeruli

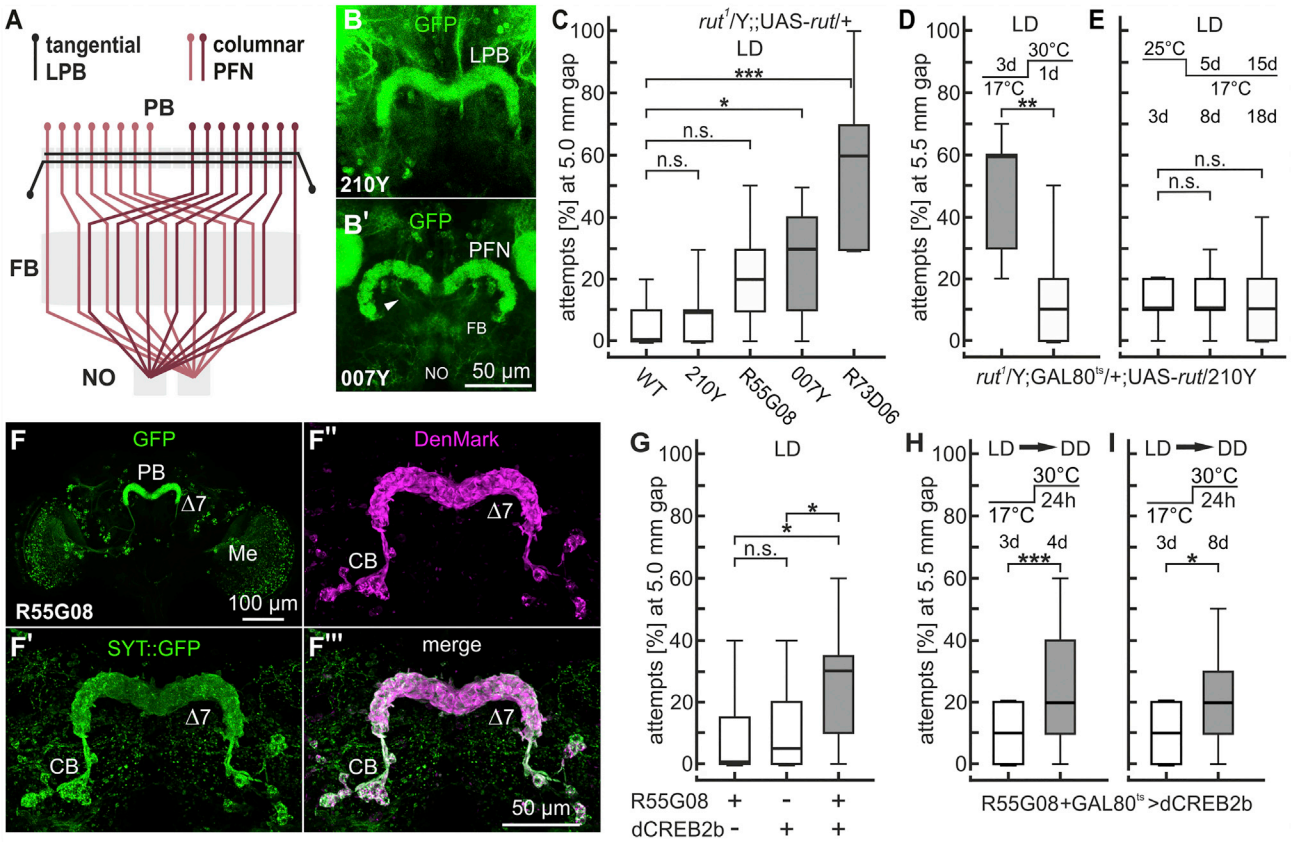

Figure 4. dCREB Is Required for Memory Consolidation and Maintenance in Intrinsic Neurons of the PB

(A) Schematic representation of the central complex depicting the tangential latero-lateral sets of protocerebral bridge (LPB) neurons and the columnar PFN fiber system linking the PB, fan-shaped body (FB), and paired noduli (NOs). 210Y- and R55G08-GAL4 drive expression in latero-lateral cells of the PB linking the neuropil with columnar local neurons.

(B) 210Y-GAL4 driver line drives tandem mCD4 expression in LPB neurons (81.75-μm z projection).

(B′) 007Y-GAL4 expresses GFP in different types of columnar PB neurons including PFNs and neurons of the w, x, y, and z bundles (arrowhead; 75.75-μm z projection).

(C) *rut*^{1} mutant rescue experiments using 210Y- and R55G08-GAL4 driver lines for LPB neurons and 007Y- and R73D06-GAL4 for PFNs reveal a requirement of *rut* in lateral Δ7 neurons (R55G08) to establish body-reach estimation. N = 15 flies per group raised in LD. Thick lines denote medians, boxes denote 25% and 75% quartiles; whiskers denote 5% and 95% quantiles; n.s., not significant; *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$; Kruskal-Wallis test and post hoc correction for multiple testing. For a comparison of small and big *rut*^{1} flies, see Figure S1 and Table S1.

(D) Conditional expression (by adding *Tub*>GAL80^{ts} to the rescue experiments) of Rut at the adult stage can rescue the body-reach estimation deficits. Flies were reared in LD at 17°C and tested when 3 days old. The same flies were then shifted to 30°C for 24 h to induce Rut expression and tested again; N = 15 flies; Wilcoxon test for dependent groups. Tests were performed at a 5.5-mm gap to compensate for the larger body size of flies raised at low temperature (applies also to E, H, and I).

(E) Flies were raised in LD and kept at the permissive temperature (25°C) for 3 days, tested, and shifted to the restrictive temperature for another 5 or 15 days. Still they were able to retrieve their body-size memory. N = 15; Wilcoxon test for dependent groups.

(F) 39.00-μm z projection of tandem mCD4::GFP expression in an adult brain under the control of the R55G08-GAL4 driver line. About 18 lateral Δ7 neurons per hemisphere innervate the PB. Some neurons in the medulla (Me) of the optic lobes are also visible. Driving the Synaptotagmin-GFP (F′) and DenMark (Icam5::mCherry; F″) fusion protein as presynaptic and postsynaptic markers, respectively, reveals that this population of lateral PB neurons together has in- and output sites in all the glomeruli of the PB (F′–F‴; 54.75-μm z projection).

(G) Expression of dominant-negative dCREB2-b in Δ7 neurons via R55G08 prevents body-size memory formation in comparison to controls. N = 20 for all groups raised in LD; Wilcoxon test for independent groups; corrections for multiple testing.

(H) Flies were raised in LD at 17°C so that they could learn their body size. When 3 days old, they were pre-tested for body-reach estimation (<30% attempts; white box) and shifted for 24 h to 30°C (in constant darkness; DD) to induce dCREB2-b (4d, gray box). See Figure S2 for temperature controls. N = 21 per group; Wilcoxon test for dependent groups.

(I) A separate group was also tested at day 3 but then remained at 17°C for another 4 days in constant darkness (DD; white box) and then shifted to the permissive temperature of 30°C and tested the next day (8d, gray box). Flies of both groups lose their body-size memory after induction of dCREB2-b. N = 21 per group; Wilcoxon test for dependent groups.

of the PB and therefore could modulate information between glomeruli and innervating columnar neurons (Figures 4F–4F′′′). This is supported by a recent study on the functional connectivity of the CX, which shows that Δ7 neurons, for instance, receive input from the EB and pass it on to different types of PFNs [29]. Whether PFNs play a role in the formation of body-size memory remains to be seen.

To exclude a developmental requirement of Rut for body-reach estimation in the adult, we employed the temperature-inducible GAL4 repressor transgene *Tub*>GAL80^{ts}, which prevents GAL4-mediated transcriptional activation in all cells [30]. *rut^{1}* rescue flies were raised in LD at the restrictive temperature of 17°C and tested at day 3. Then, we shifted the same flies to 30°C overnight to induce cDNA expression before testing them again the next day. Paired comparison of individual flies showed that Rut expression in the adult is sufficient for proper body-reach estimation (Figure 4D). Moreover, it shows that this small group of intrinsic LPB neurons (∼18 per hemisphere) remains plastic until they have the right amount of adenylyl cyclase to learn their body size, supporting the finding that there is no "imprinting period" for body-size memory. On the other hand, when we suppressed Rut expression in the LPB neurons of *rut^{1}* flies only after memory formation while aging (LD cultured) the memory remained stable, indicating that cAMP signaling is not required for retrieval or long-term memory maintenance (Figure 4E).

Body-Size Memory Consolidation and Maintenance Require dCREB2

cAMP-PKA (protein kinase A) signaling is required for certain types of short-term memories [21], but also for long-term memory formation in olfactory associative learning via activation of the transcriptional regulator dCREB2 [12, 31–33]. Therefore, we expressed a dominant-negative form of dCREB2 (R55G08>UAS-dCREB2-b [34]) in the Δ7 neurons to assess whether body-size memory also requires dCREB2 activity. In agreement with our hypothesis, expression of the dCREB2 repressor in these LPB neurons resulted in defective body-reach estimation (Figure 4G). Next, we asked whether flies that have acquired a body-size memory need continuous dCREB2 activity to maintain their memory later on. Therefore, flies that can express dominant-negative dCREB2-b in Δ7 neurons under the control of R55G08-GAL4 and *Tub*>GAL80^{ts} were reared in LD at the restrictive temperature of 17°C. When 3 days old, these flies were tested as to whether they had acquired a body-size memory, and were then shifted to darkness and the permissive temperature of 30°C for 24 h. When we tested these flies again, they had nearly lost their body-size memory, indicating that this type of memory requires dCREB2 activity after consolidation (Figure 4H).

It is noteworthy that analysis of aversive olfactory long-term memory in the Kenyon cells of the mushroom body revealed that the requirement for dCREB2 activity ceases after 4 days [35], and therefore we extended our experiment. 3-day-old flies (LD at 17°C) were pretested and then set at 17°C in constant darkness (DD) for another 3 days. On day 7, they were shifted to the permissive temperature of 30°C for 24 h to induce dCREB2-b in aged flies. Testing these now 8-day-old flies resulted also in increased climbing attempts, showing that the body-size memory is significantly reduced (Figure 4I). Note that this is not due to the elevated temperature, because R55G08-GAL4 *Tub*>GAL80^{ts} control flies were unaffected (Figure S2). The finding suggests that this long-lasting memory depends on the continuous expression of dCREB2 target genes for memory maintenance.

Conclusions

Knowledge about one's own body size is equally important to prey and predator, establishes hierarchies among competitors, and is essential to navigate through complex environments. However, how insects gain this knowledge and store it in a long-term memory is largely unknown. Ben-Nun et al. [36] showed that hemimetabolous locusts have to learn their new body size and reach after the last mold. Newly emerged adults try to climb the same maximum gap sizes that they managed as Vth instar larvae. However, spending 3 h (or more) in a "gap-enriched environment" suffices to learn their new, larger, body size [36]. In our experiments, trained flies needed several hours (∼12) to recall their body size. This difference might be due to the fact that locusts use visual and antennal mechanosensory inputs to climb gaps [37], and suggests that this dual input facilitates learning and memory formation. Similar to our results in flies, inhibition of protein synthesis at the time of training prevented adult locusts from learning their new body size [36]. Whether this memory is as stable and enduring as body-size memory in flies, which lasts at least for 21 days, remains to be seen.

Comparing *Drosophila* body-size memory to other, associative, long-term memories that are built in the mushroom body for punished [14] and rewarded odors [38, 39] or courtship suppression [40] that last only about 1 week revealed interesting differences. In contrast to olfactory memories, which are instantaneously available [18], the learned body size cannot be retrieved for several hours, although it seems to be consolidated after 2 h. Like consolidation of these other long-term memories, body-size memory requires the transcriptional activator dCREB2 [31, 33, 41]; however, this requirement seems to be ongoing to maintain the memory stable.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at https://doi.org/10.1016/j.cub.2019.04.037.

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AUTHOR CONTRIBUTIONS

T.K. and L.S. performed all experiments and statistical analyses. T.K., L.S., B.P., and R.S. prepared the figures. B.P. and R.S. designed the study and wrote the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
chicken polyclonal anti-GFP	Aves Labs	Cat# GFP-1020; RRID:AB_10000240
goat polyclonal anti-mCherry	SICGEN	Cat# AB0040-200; RRID:AB_2333092
goat anti-chicken IgG (H+L) Alexa Fluor 488 conjugated	Molecular Probes	Cat# A-11039; RRID:AB_142924
donkey anti-goat IgG (H+L) Cy3 conjugated	Jackson ImmunoResearch Labs	Cat# 705-165-147; RRID:AB_2307351
Chemicals, Peptides, and Recombinant Proteins		
Vectashield Mounting Medium	Linaris	Cat# H-1000; RRID:AB_2336789
Horse Normal Serum	Linaris	Cat# S-2000; RRID:AB_2336617
Cycloheximide	PanReac AppliChem	Cat# A0879,0005
Patent blue V sodium salt	Sigma	Cat# 21605
Sigmacote	Sigma	Cat# SL2
Experimental Models: Organisms/Strains		
wild-type Canton-S (CS)	Bloomington *Drosophila* Stock Center	RRID:BDSC_64349
oc [1] (*ocelliless*)	[42]	FBal0013199
rut [1] (*rutabaga*)	Bloomington *Drosophila* Stock Center	RRID:BDSC_9404
dnc [1] (*dunce*)	Bloomington *Drosophila* Stock Center	RRID:BDSC_6020
w [*];; 007Y-GAL4	[23]	RRID:BDSC_30812
w [*];; 210Y-GAL4	[23]	FBti0004605
w [1118];; R55G08-GAL4	[25]	RRID:BDSC_50422
w [1118];; R73D06-GAL4	[25]	RRID:BDSC_46692
w [1118];; P(UAS-*rut.Z*)1	[22]	FBti0038654
P(UAS-dCREB2-b)	[34]	FBtp0052796
y [1] *w*[*]; P(UAS-CD4-tdGFP)8M2	Bloomington *Drosophila* Stock Center	RRID:BDSC_35839
w [1118]; P(UAS-DenMark)2, P(UAS-syt.eGFP)2; In(3L)D/TM6C,*Sb* [1]	Bloomington *Drosophila* Stock Center	RRID:BDSC_33064
w [1118]; L [1]/*CyO*; P(UAS-DenMark)3, P(UAS-syt.eGFP)3	Bloomington *Drosophila* Stock Center	RRID:BDSC_33065
w [*]; *sna*[Sco]/*CyO*; P(TubP-GAL80[ts])7	Bloomington *Drosophila* Stock Center	RRID:BDSC_7018
R55G08-GAL4 P(TubP-GAL80[ts]/TM3,*Sb*	this study	N/A
Software and Algorithms		
Statistica 8.0	StatSoft	http://www.statsoft.de/
R version 3.4.3 (2017-11-30)	StatSoft	https://www.r-project.org/
Leica Application Suite X	Leica Microsystems	http://www.leica-microsystems.com/products/microscope-software/details/product/leica-las-x-ls/ /
Fiji	NIH [43]	http://fiji.sc/
CorelDRAW Graphics Suite X7 Version 17.1.0.572 (2014)	Graphic Design Software	https://www.coreldraw.com/de/
Other		
Leica TCS SP8 microscope	Leica Microsystems	http://www.leica-microsystems.com/products/confocal-microscopes/details/product/leica-tcs-sp8/
Zeiss OPMI 1-F dissection microscope	Zeiss	N/A
Picasso CRT Image Synthesizer	Innisfree, Redwood City, USA	N/A
2 Tektronix 608 oscilloscope screens	Tektronix, Beaverton, USA	N/A

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Roland Strauss (rstrauss@uni-mainz.de).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Drosophila melanogaster strains were maintained on standard medium (water, cornmeal, malt extract, molasses, yeast, soybean, agar, preservative methyl-4-hydroxybenzoate) at 25°C, 60% humidity, and 14h / 10h light-dark cycles. If not noted otherwise, 3-day-old males were used with their wings cut to one-third of their length under cold anesthesia one day prior to the experiment. Wings of dark-reared flies were clipped under red light invisible to flies. Wild-type Canton-S (CS) flies served as control in all experiments or were used to generate heterozygous flies. GAL4 driver lines 007Y and 210Y were provided by D. Armstrong. oc^1 is from the former collection of K.G. Götz, and the effector line UAS-dCREB2-b was provided by R. Davis. The conditional diver line R55G08-GAL4 *Tub* > $GAL80^{ts}$ was created through meiotic recombination.

For experiments presented in Figures 1 and S1 offsprings from the same vial were visually sorted for especially small or big flies and separated into two groups. After the behavioral test, the body size was characterized by measuring the lengths of tibia plus tarsi of the front legs for each fly *post mortem* (Table S1). Legs were cut at the femur, flattened out and fixed with double-faced adhesive tape on glass slides and video imaged under a stereomicroscope.

METHOD DETAILS

Behavioral Analysis

A black polycarbonate plastic block (34 × 10 × 4 mm^3, see Figure 1) with a 5-mm-deep gap in the middle was placed on an inverted Petri dish of 44 mm diameter in the center of a plastic Petri dish of 88 mm in diameter filled with water. The water moat and clipped wings prevented escape. A white cardboard cylinder around the larger dish shielded the fly from external visual stimuli. Ten approaches to the gap were observed from the side through a dissection microscope (Zeiss OPMI 1-F, Schott cold light KL1500). The behavior was recorded by direct observation in the following categories: turning around (Video S1), walking down (Video S2), falling down, successful or unsuccessful cross-overs (videos S1 and S2 in [3]); most importantly, the occurrence of leg-over-head behavior was noted for each approach to the gap. The latter is a unique sign for a crossing attempt at a gap that cannot be easily surmounted by a large step. The occurrence has been counted in 10 approaches to the gap per fly and gives the percentage of attempts shown in the figures. Gap size was 5.0 mm in most of the experiments except in Figures 1C–1F (2.0 mm to 6.0 mm); it is indicated in each panel. Flies that were raised at the low temperature of 17°C were tested at a 5.5 mm gap to compensate for the larger body size.

Dark-reared (DD) flies were cultured in the constant dark incubator and their wings were cut under red light (> 600 nm), invisible to flies. To assess their naive state, flies were taken from the dark, immediately tested (max. 3 min in ambient light) and returned to the dark to be retested on the next day. For manipulation of the parallax motion vision system, naive (DD) flies were trained for 10 min in a LED arena of 40 cm diameter ([44] luminance 30 cd m^{-2}, contrast 0.94; Figure 2E). The walls of training vials of 20 mm diameter were coated with SigmaCote to keep flies on the bottom and to prevent them from experiencing additional views of the stripped patterns. One vial at a time with one to three flies was centered in the middle of the arena. For manipulation of the experienced parallax motion, single naive DD flies were walking for 15 min on a seesaw (tilt ± 10°) that was manually flipped whenever the fly had reached the upper end (Figure 2E′; Video S3). Negative geotaxis is the drive for ascending again. The parallax motion that walking flies perceived from dark and bright vertical stripes on two flat screens facing each other, was partially compensated by letting the patterns drift in the direction of walking at half their average walking speed; i.e., 7 mm s^{-1} (Picasso CRT Image Synthesizer and two oscilloscope screens Tektronix 608; pattern wavelength 24°, height 105 mm, width 115 mm; distance screen to seesaw 100 mm; manual flip of drift direction whenever the fly reversed its walking direction). The pattern was manually stopped whenever the fly stopped. Flies, which ceased walking before the end of the training, were excluded from the experiment.

Cycloheximide Treatment

Flies were fed 200 μl 35 mM cycloheximide dissolved in 10% glucose/3% ethanol on filter paper for 2 h at room temperature. To prove drug intake, Patent blue V sodium salt was added to the solutions causing a blue staining of the fly abdomen. After treatment, flies were replaced in vials containing food until testing the next day.

Immunohistochemistry

Adult brains were dissected in ice-cold PBS and fixed in 4% paraformaldehyde/PBS for 30 min at room temperature and for another 30 min in 4% paraformaldehyde/PBS/0.2%Trition-X100. All following washes were done in PBS/0.5%Triton; (except for a permeabilization step for 1 h in PBS/1%Trition at room temperature). Specimens were blocked for at least 1 h in 4% normal horse serum PBS/0.5%Triton at room temperature. Primary antibodies (α-GFP 1:1000 and α-mCherry 1:200) were incubated for at least 36 h in 4% serum/PBS/0.5%Triton at 4°C. Secondary antisera (goat α-chicken/donkey α-goat) were incubated for at least 36 h at 4°C

in PBS/0.5%Triton at a dilution of 1:1000. Brains were cleared in 50% glycerol/PBS at least overnight at 4°C and mounted in Vectashield. Images were acquired using a Leica TCS SP8 confocal microscope and processed with LASAF (Leica) and Fiji [43]. CorelDraw was used to assemble the micrographs.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistics

Normality was tested with Kolmogorov-Smirnov-, and Lilliefors-, or, Shapiro Wilk's W test. For simple comparisons, normally distributed sets of data were compared with the Student t test. If one or both samples were not normally distributed, the Wilcoxon-test was applied. For dependent variables, either t test for dependent samples or Wilcoxon Matched Pairs Test was used, respectively. For all multiple-group comparisons, post hoc corrections were made. Statistical analyses were performed with STATISTICA 7.0 or R 3.4.3 software. Thick lines in boxes represent the median, boxes the 25% and 75% quartiles, whiskers the 5% and 95% quartiles. ns: not significant; *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$ in all figures.

DATA AND SOFTWARE AVAILABILITY

Raw data are available on reasonable request from the corresponding author. All software used is commercially available with the exception of R, which is freely available under the GNU General Public License.